

App Libraries

Library	Role in this app	Why it's the right tool	Category
streamlit	Renders the web page itself — the form, buttons, and result display	Purpose-built for exactly this: turning a Python script into a usable app with minimal code	Application framework
pandas	Organizes the user's input into a table (DataFrame) the model can read	The model was trained on pandas DataFrames, so predictions must be shaped the same way	Data handling
numpy	Supports the underlying numeric operations pandas and the model rely on	Standard numerical foundation for almost all Python data science tools	Numerical computing
joblib	Loads the saved model, encoders, and scaler back into memory	The standard, efficient way to save and reload trained scikit-learn / XGBoost objects	Model persistence